

HARSHANA WELIGAMPOLA

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SUMMARY

PhD candidate in Electrical and Computer Engineering with over 4 years of academic research experience. Demonstrated expertise in advanced computational imaging systems, generative AI models, and optics. Collaborated effectively in cross-functional teams to innovate and develop prototypes, influencing product development strategies. Committed to bringing cutting-edge engineering solutions and analytical acumen to dynamic research environments.

EDUCATION

PhD: Electrical and Computer Engineering

Purdue University, USA

May 2027 (Expected)

GPA 3.97

BSc: Computer Engineering

University of Peradeniya, Sri Lanka

Dec 2020

GPA 3.95

SKILLS

Programming Languages: Python, C, Java, C++, MATLAB

Simulation: Zemax OpticStudio

Deep Learning Frameworks: Tensorflow, PyTorch

Hardware-oriented Programming: Arduino, Verilog HDL, Assembly

RESEARCH EXPERIENCE

MetaZoom; Compact high-zoom metasurface system

Purdue University | West Lafayette, USA.

Aug 2022 - Aug 2024

- Configure a metasurface optical system for high-zoom image capture applications, ensuring optimal performance at miniature scales
- Achieved the best telephoto ratio of 0.44 by reconstructing high-quality images from the innovative optical setup
- Enhanced the optical system's effectiveness by 30%, leveraging advanced metasurface design analysis on refractive optics
- Mitigated image aberrations through the development of a sophisticated post-processing pipeline, improving image clarity.

Computer vision for low light images

Sri Lanka Technological Campus | Padukka, Sri Lanka

Aug 2020 - Apr 2021

- Conducted extensive research on computer vision techniques addressing challenges in processing and enhancing low-light images.
- Improved low-light image clarity using machine learning algorithms, achieving a 25% increase in visibility and contrast through advanced image enhancement

Study on Deep Learning for Latency Constraint Applications in Wireless Systems

Nokia Bell Labs | Antwerp, Belgium

Feb 2019 - Aug 2019

- Evaluated the effectiveness of using deep learning approaches under time-constrained communication applications.
- Enhanced communication efficiency by implementing deep learning models, reducing latency by 23%.

AWARDS & HONORS

Ross Fellowship

Purdue University

Jan 2022

- Received the Ross Fellowship from Purdue University, enhancing academic distinction by funding graduate studies and enabling the presentation of innovative research at national conferences.

1st Country Rank and 62nd World Rank, IEEEExtreme 10.0 (out of 3000+ teams)

IEEE

Jan 2016

- Achieved 1st Country Rank and 62nd World Rank among over 3,000 teams in IEEEExtreme 10.0, demonstrating exceptional problem-solving skills and competitive programming expertise

Bronze Medal, Mumbai, India.

International Physics Olympiad

Jan 2015

- Secured the Bronze Medal at the International Physics Olympiad in Mumbai, India, demonstrating problem-solving skills and academic excellence among over 400 top global students.

National rank 28, (out of 60,000+)

GCE Advanced level

Jan 2014

- Secured the 28th national rank among over 60,000 participants in the GCE Advanced Level examinations, demonstrating exceptional academic performance and subject mastery.

PUBLICATIONS

- Harshana Weligampola, Yuanrui Chen, Abhiram Gnanasambandam, Dilshan Godaliyadda, Hamid R. Sheikh, Stanley H. Chan, and Qi Guo, "MetaTele: compact refractive metasurface computational telephoto camera," Accepted for Optics. Express, 2026
- Harshana Weligampola, Gihan Jayatilaka, Suren Sritharan, Parakrama Ekanayake, Roshan Ragel, Vijitha Herath, Roshan Godaliyadda, "An optical physics inspired CNN approach for intrinsic image decomposition", Accepted for IEEE International Conference on Image Processing (IEEE ICIP), 2021
- Suren Sritharan, Harshana Weligampola, Haris Gaćanin, "A Study on Deep Learning for Latency Constraint Applications in Beyond 5G Wireless Systems", Accepted for IEEE Access, 2020